Claim-Eligibility Auditing for Post-hoc Explanations: Synthetic Calibration and Cross-Domain Cases

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Abstract—**Post-hoc explanations may be faithful to a fitted model even when that model has not shown enough evidence to support a claim about locked outcomes. We introduce a claim-eligibility audit that checks overall predictive adequacy, incremental feature-group value, and event recovery before progressively stronger interpretations are allowed. In synthetic experiments, ungated explanation permits all 240 structurally invalid runs; the audit reduces false permission to 171 while retaining 88.9% sensitivity. The remaining errors occur mainly in leakage, confounding, and distribution-shift regimes that preserve predictive performance, showing that predictive checks cannot replace study-design validation. On a locked U.S. crop panel, the selected Weather-only ExtraTrees model does not pass any of the three required modules. A local SHAP explanation is internally coherent, yet the model predicts $+0.209$ for an event whose observed residual is -0.510, so the explanation is retained only as a description of the fitted model. A county analysis provides incremental feature evidence but does not pass overall adequacy, whereas a PJM demand case passes the applicable modules. Together, these cases show how the same audit can restrict, abstain from, or permit post-hoc interpretation.**

Index Terms—**claim eligibility, explainable machine learning, temporal validation, crop yield, reproducibility**

I. Introduction

Post-hoc explanations are often read as soon as a fitted model produces an interesting attribution. That habit is risky when the model has not shown that it can support the requested study-level claim on future or otherwise locked outcomes. A coherent attribution may describe the model's calculation while the model still misses the observed event.

Faithfulness, stability, and robustness evaluate the relation between an explanation and a fitted predictor. Claim eligibility asks a prior question: whether the predictor has enough locked evidence to support the interpretation being requested. The distinction matters because a faithful explanation of a weak or misdirected predictor should not become evidence about weather damage, feature reliance, or event recovery.

We evaluate that prior question with a claim-eligibility audit. The audit uses locked, same-row comparisons with paired uncertainty to test overall predictive adequacy (Module A), incremental feature-group value (Module B), and event recovery (Module E). The result is not a row-level reject option, as in selective prediction, but a study-level permission rule for how far post-hoc explanations may be interpreted.

We evaluate the audit in two complementary settings. Synthetic regimes with known claim labels quantify false permission and false abstention, while crop, county, and PJM cases show how the same rule produces different claim-level decisions. The state crop panel abstains, the county case has incremental feature evidence but does not pass overall adequacy, and the PJM case permits predictive-reliance interpretation. Each verdict is tied to the locked rows, target construction, prediction vectors, uncertainty estimates, and configuration that produced it. This combination of calibration, contrasting empirical outcomes, and traceability is the paper's contribution.

II. Related Work

A. Explanation Methods and Their Boundary

SHAP and LIME explain a model output under their respective approximation and reference choices [1], [2]. Permutation-style analyses, model reliance, accumulated local effects, and conditional importance likewise describe feature dependence or behavior of a fitted predictor [3]–[5]. They are useful diagnostic artifacts, but they do not by themselves show that a model can support an outcome claim on locked rows. Prior work also shows that post-hoc explanations can be sensitive to perturbation behavior, model randomization, deletion/retraining benchmarks, and the causal status of the target estimand and reference distribution [6]–[10]. Recent surveys, removal-based formulations, counterfactual critiques, and explanation benchmark suites further emphasize that faithfulness and robustness must be evaluated against explicit tasks rather than assumed from visual plausibility [11]–[15].

B. Selective Prediction and Abstention

Selective classification and reject-option systems trade coverage for risk: a predictor may abstain when its risk target is not satisfied [16], [17]. Our unit is different. The decision is not a per-row class label but a study-level permission to interpret a fitted feature family at a specified claim level. The protocol therefore couples a predictive-adequacy module with a paired feature-group value module and reports false permission and false abstention on synthetic scenarios with known data-generating mechanisms and pre-specified evaluation labels.

C. Evaluation, Leakage, and Reproducibility

Random splits can be inappropriate when temporal or hierarchical structure is present [18]. Leakage can create seemingly strong scientific machine-learning results while invalidating the intended forecast claim [19]. Reproducible ML also requires explicit code, data paths, settings, and numerical evidence [20]. We therefore use locked temporal evaluation and retain the configuration, prediction vectors, bootstrap draws, and table inputs needed to recompute every reported comparison.

D. Yield Modeling and Detrending

Detrending separates a local historical yield trajectory from residual variation, but the result can depend on the detrending procedure and information available at the evaluation date [21], [22]. We therefore fit each crop–state trend and residual scale on training observations only. Ridge regression, Random Forest, and ExtraTrees provide deliberately distinct linear and tree-based model families [23]–[25]; the implementations use scikit-learn [26]. The crop panel is the main application context, not the source of the audit logic.

III. Data and Protocol

A. Panel, Features, and Target

The unit is a crop–state–year row. The processed panel contains 1,257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states: Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, and Washington. Yield in t ha^{-1} comes from USDA NASS Quick Stats [27]; daily maximum/minimum temperature, precipitation, and shortwave radiation come from NASA POWER [28]. The primary design uses 35 full-season weather features, so the task is a post-season scientific audit of residual-model reliance rather than a pre-harvest forecast.

All targets are constructed from training-period information only, so later yields cannot alter previously defined evaluation rows. For a test interval, each crop–state linear trend and residual scale is estimated using only its associated training rows. The learned target is the raw train-only residual in t ha^{-1}: $\hat{y}_{c,s,t} = a^{(\mathrm{train})}_{c,s} + b^{(\mathrm{train})}_{c,s} t$, $r_{c,s,t} = y_{c,s,t} - \hat{y}_{c,s,t}$, and $z_{c,s,t} = r_{c,s,t}/\max(\hat{\sigma}_{c,s,\mathrm{train}}, \epsilon)$. The standardized residual defines events, $\mathbb{1}[z_{c,s,t} < \tau]$, only; raw residuals remain the prediction target. This distinction prevents a pooled raw-error score from being read as a standardized event-recovery metric.

A row is eligible only with at least three training observations. The validation period begins with 144 candidate rows and retains 140 after 4 pre-specified exclusions. The locked test starts with 343 candidates, excludes 10, and evaluates 333 rows. Min-history rules of 3, 5, 8, and 10 years are reported as sensitivity analyses because both eligibility and the fitted residual scale can change with available history. Three observations are the minimum that leaves one positive residual degree of freedom for estimating a training residual scale.

The model matrices exclude year, yield, trend, residual, standardized residual, event labels, predictions, history length, and residual scale. A separate sensitivity design adds early-, middle-, and late-season proxies without redefining the target or locked population. Feature definitions, forbidden-column checks, and overlap audits are consolidated in the reproducibility record.

B. Selection and Baselines

Ridge $\alpha \in \{1, 10\}$ and Random Forest/ExtraTrees with 160 trees and $\mathrm{min\_samples\_leaf} \in \{1, 2\}$ are evaluated on metadata-only, weather-only, and full representations. Stochastic fits use fixed seeds $\{7, 17, 27, 37, 47\}$; numeric imputation/scaling, categorical encoding, and model fitting occur within the training fold. Mean per-row seed-aggregated validation RMSE chooses the configuration, with lexicographic tie-breaking and no locked-test access. Seed-level RMSE standard deviations are reported so tiny differences are not presented as deterministic wins.

The final audit compares learned predictions with baselines on the same locked rows, so each difference is a paired comparison rather than a difference between separate evaluation populations. The baselines are zero residual, training-mean residual, and crop training-mean residual. We use 2,000 deterministic year-block bootstrap replicates and percentile 95% intervals over the ten locked year-block units. With only ten locked year-block units, interval precision and statistical power are necessarily limited. We nevertheless retain year-block paired resampling because it preserves the cross-sectional dependence shared by crop–state rows within the same calendar year. We cite stationary bootstrap for the dependent-resampling motivation but resample complete calendar years as paired blocks [29]. Prediction hashes, pairwise vector differences, and the state-year and crop–state cluster sensitivity schemes are retained in the reproducibility record.

C. Pre-specified Claim Modules

Module A asks whether the validation-selected model improves on a prespecified baseline over the locked population. Module B asks whether the requested feature family adds predictive value beyond metadata on the same rows. Module E is invoked only for event-level claims and asks whether the model recovers the prespecified tail in error magnitude, ranking, and top-k prioritization. These questions define the highest claim level that the evidence can support.

Let P^* denote the locked target population implied by the split, target construction, feature set, and evaluation period. Module A estimates $\theta_A = \mathrm{RMSE}_{P^*}(f^*) - \mathrm{RMSE}_{P^*}(f_0)$, where f^* is the validation-selected model and f_0 is the zero-residual baseline. Module B estimates $\theta_B = \mathrm{RMSE}_{P^*}(f_{M+W}) - \mathrm{RMSE}_{P^*}(f_M)$, the incremental weather value beyond metadata. The reported estimates replace P^* by the locked empirical rows, use paired year-block resampling, and pass only when the upper 95% interval is below zero.

Module E is invoked only for below-trend event claims. Its primary tail is $z < -1$; $z < -1.5$ and $z < -2$ are sensitivities because their smaller samples reduce power. The event module requires tail RMSE/MAE improvement, rank recovery with a positive lower bootstrap bound and within-year permutation $p < .05$, and top-k recovery above chance by lift, hypergeometric probability, permutation p, and year-block lift interval. Module D is an outside-path Weather-only versus Metadata-only representation diagnostic and does not affect claim permission. In this crop panel it is numerically identical

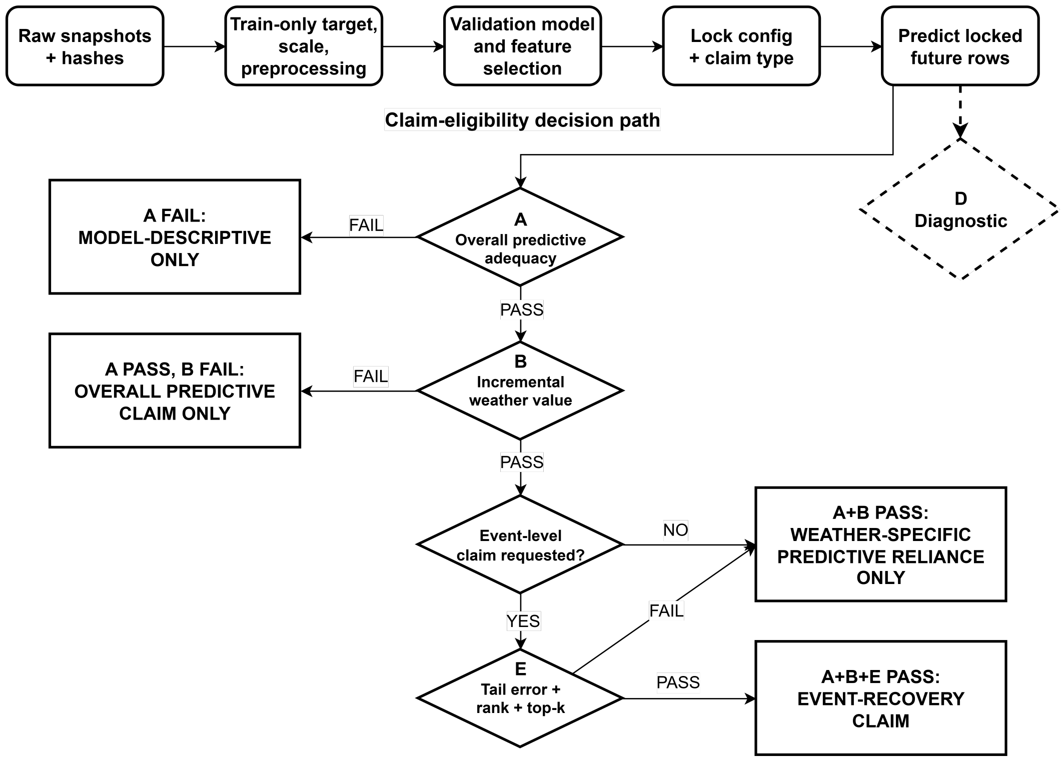

Fig. 1. Sequential claim-eligibility decision path. Module A controls whether interpretation may extend beyond model description, Module B controls incremental weather-specific predictive reliance, and Module E is evaluated only when an event-level claim is requested. Module D remains diagnostic only and does not affect claim permission.

to Module A because Metadata-only predictions collapse to the zero-residual baseline; detailed prediction-vector equality checks are retained in the reproducibility record. The fixed claim-eligibility audit is summarized in Table I.

IV. Results

A. Validation and Locked Test

On the locked 2016–2025 panel, the selected Weather-only ExtraTrees model does not improve reliably on the zero-residual baseline: ΔRMSE is -0.005 with a 95% CI of $[-0.019, 0.009]$. Module A therefore does not pass. The Full-minus-Metadata contrast is also uncertain (ΔRMSE $= -0.012$, 95% CI $[-0.029, 0.002]$), so Module B does not pass. The selected configuration had the lowest mean validation RMSE, but its advantage was small relative to seed variability and did not persist from the 2012–2015 validation period (RMSE 0.384) to the 2016–2025 locked period (RMSE 0.669).

The selected validation configuration is ExtraTrees with Weather-only features and locked $R^2 = -0.014$. Module A uses this single validation-selected configuration. Module B is a fixed architecture contrast: Full and Metadata-only predictions reuse the Module A configuration ID and swap only the feature family. The selected weather-only ExtraTrees configuration ranks first in 3/5 stochastic validation seeds, so we retain the pre-specified selection without claiming clear superiority. The grid, tie rule, seed-level scores, locked-test access flag, and prediction hashes are retained in the selection record. Table II reports all same-row locked comparisons.

B. Event-Detection Diagnostics

Event detection is evaluated over all 333 locked rows, not only a tail preselected by the observed outcome. The score is negative predicted residual and the score threshold is fixed at zero before final evaluation. Event scores retain weak global discrimination: ROC-AUC is 0.641 for the primary $z < -1$ tail and 0.672 for the two severe-tail sensitivities; primary-tail PR-AUC is 0.338 versus prevalence 0.219. This does not contradict the module decision. ROC-AUC summarizes pairwise ordering over all locked rows, whereas Module E asks whether the model improves paired error and recovers the pre-specified tail beyond rank and top-k null checks. The available discrimination is diagnostically positive but insufficient for the requested observed-event claim.

C. Below-Trend Event Module and Robustness

The broad below-trend subset contains 73 locked-test rows. The primary tail error contrast is favorable but uncertain: tail ΔRMSE is -0.043 with a 95% CI of $[-0.074, 0.007]$. Rank uncertainty crosses zero, and the primary $z < -1$ top-10 result recovers 1/10 events, below random expectation and without a significant permutation result. Module E therefore does not pass. The components target distinct evidentiary requirements: tail RMSE/MAE tests error magnitude, rank statistics test ordering, and chance-adjusted top-k checks test prioritization

TABLE I

FIXED CLAIM-ELIGIBILITY MODULES AND LOCKED CROP-PANEL VERDICTS. MODULE D (WEATHER-ONLY VERSUS METADATA-ONLY) IS A DESCRIPTIVE REPRESENTATION DIAGNOSTIC OUTSIDE THE PERMISSION PATH AND IS THEREFORE NOT A REQUIRED ROW. FULL CONFIGURATION DETAILS ARE IN THE REPRODUCIBILITY MANIFEST.

Mod.	Question/contrast	Pass rule	Current estimate	Verdict	Claim consequence
A	Selected Weather-only vs zero	upper paired CI < 0	-0.005 [-0.019, 0.009]	DOES NOT PASS	model description only
B	Full vs Metadata-only	upper paired CI < 0	-0.012 [-0.029, 0.002]	DOES NOT PASS	no weather-specific reliance claim
E	Primary tail error + rank + top-k	all checks pass	RMSE -0.043 [-0.074, 0.007]; rank 0.180; top-10 1/10	DOES NOT PASS	no event-recovery claim

TABLE II

LOCKED 2016–2025 SAME-TASK AUDIT. LEARNED-MODEL PREDICTIONS ARE SEED AGGREGATED. ΔRMSE IS MODEL MINUS ZERO-RESIDUAL RMSE WITH PAIRED 95% YEAR-BLOCK INTERVALS; RMSE IS T HA^{-1}. THE MODULE B CONTRAST IS COMPUTED DIRECTLY AS FULL-MINUS-METADATA; IN THIS PANEL IT MATCHES FULL-MINUS-ZERO BECAUSE METADATA-ONLY PREDICTIONS COINCIDE WITH THE ZERO-RESIDUAL VECTOR TO FLOATING-POINT PRECISION.

Model	Features	n	Seeds	R^2	RMSE	ΔRMSE vs zero [95% CI]
Crop train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Zero residual	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Full	333	5	0.006	0.662	-0.012 [-0.029, 0.002]
ExtraTrees	Metadata-only	333	5	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Weather-only	333	5	-0.014	0.669	-0.005 [-0.019, 0.009]

of the most important events. The module is not mechanically impossible to pass: in the synthetic benchmark, moderate-signal, strong-signal, and valid imbalanced-tail regimes pass Module E in 30/30 seeds. Because Modules A, B, and E do not pass, the local and global explanation outputs remain model diagnostics rather than support for observed-event or weather-specific predictive claims.

For the Barley–Colorado 2016 case, the locked Weather-only model predicts a residual of $+0.209$ while the observed residual is -0.510. The grouped SHAP decomposition is internally coherent: the unrounded weather-family contributions and remainder reconstruct the fitted prediction from its base value. However, the fitted prediction misses the observed event in both direction and magnitude. Ungated interpretation would permit a weather-attribution claim from this coherent explanation, whereas the claim-eligibility audit blocks that interpretation because Modules A, B, and E do not pass on locked outcomes.

D. Sensitivity and Traceability Summary

Sensitivity analyses varied four parts of the design: minimum history, target construction and detrending, model family, and dependence assumptions. Additional checks examined temporal prefixes and state-level heterogeneity. Some alternatives improved local metrics, but none changed the primary claim because they altered the evaluation population or did not pass another required module. The complete audit record contains 11 pre-specified sensitivity categories and 51 outputs. Figure 4 shows descriptive state-level heterogeneity; these subgroup contrasts do not replace the panel-level Module A decision.

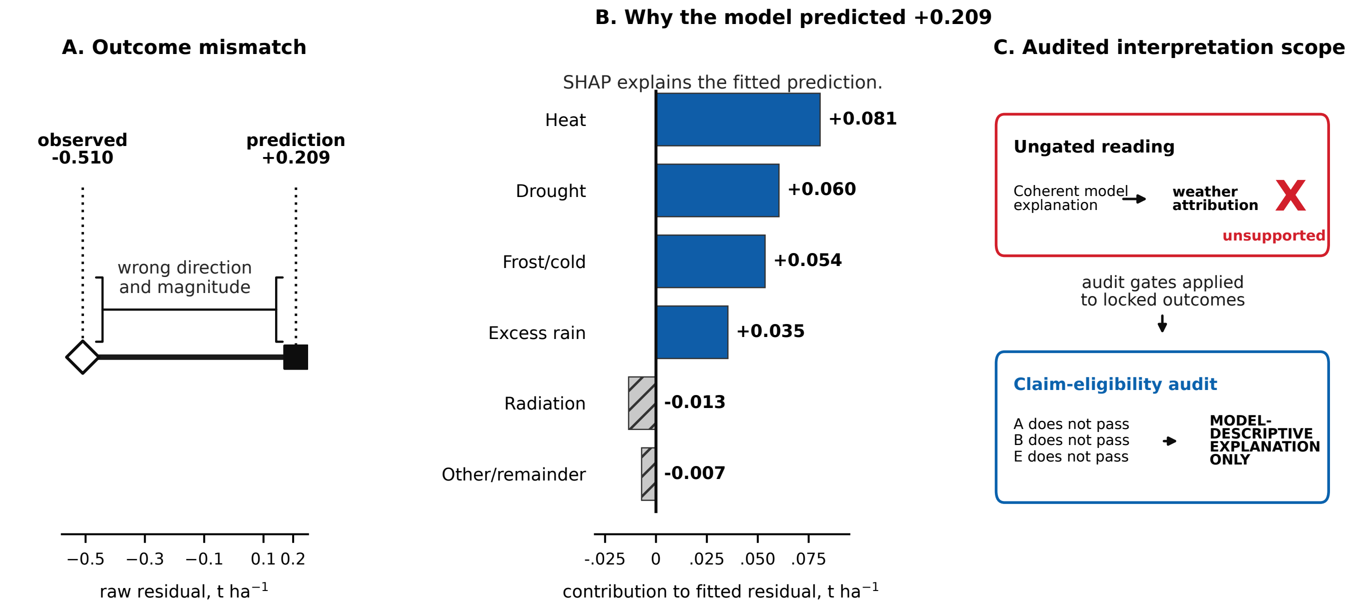

Fig. 2. A coherent model explanation does not establish an observed-event claim. (A) The locked Weather-only model predicts $+0.209$ for the Barley–Colorado 2016 row, whereas the observed residual is -0.510 t ha^{-1}. (B) Grouped SHAP contributions describe why the fitted model produced $+0.209$; the unrounded terms reconstruct the prediction and displayed values are rounded. (C) Ungated interpretation would promote the explanation to weather attribution, while the claim-eligibility audit retains only a model-descriptive explanation because Modules A, B, and E do not pass.

V. Calibration and External Cases

A. Synthetic Benchmark

The claim-eligibility audit is also evaluated where the mechanism and evaluation label are known before scoring. A repeated synthetic temporal benchmark covers 14 regimes over 30 seeds each, including correlated features, omitted confounding, temporal and geographic shift, and leakage. A regime is labeled permissible when the requested feature-group or event claim is present in the data-generating process and remains conceptually valid, even if estimation is difficult because of limited sample size or representation mismatch. A regime is labeled impermissible when the requested signal is absent or when the data-generating or study-design mechanism makes that interpretation structurally invalid. These ground-truth labels are used only to score the observable audit and are never inputs to Modules A, B, or E.

The audit behaves differently across two classes of invalid regimes. It sharply reduces permission when the requested signal is absent or substantially degraded, as in no-signal, weak-signal, and measurement-error settings. It changes little when the invalidity preserves predictive performance, as in leakage, omitted confounding, correlated features, and geographic shift.

Across all invalid runs, ungated explanations falsely permit 240 of 240 invalid runs (100.0%, 95% CI [98.4,100.0]), whereas the observable A+B+E rule falsely permits 171/240 invalid runs (71.2%, 95% CI [65.2,76.6]). The rule falsely abstains on 20/180 valid runs (11.1%, 95% CI [7.3,16.5]), with permission rate 78.8%, sensitivity 88.9%, and specificity 28.7%. The 28.7% specificity therefore reflects a specific limitation rather than a uniform error pattern: predictive evidence can discipline claims when signal is weak, but it cannot certify design validity when leakage, confounding, or shift leaves predictive performance intact.

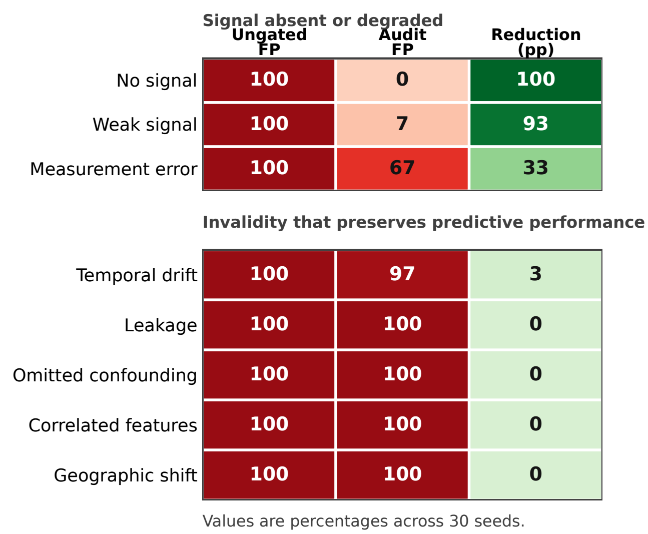

Fig. 3. False-permission patterns across structurally invalid synthetic regimes. The audit sharply reduces permission when the requested signal is absent or degraded, but changes little when leakage, confounding, correlated features, or shift preserve predictive performance. Values are percentages across 30 seeds; ground-truth labels are used only for evaluation and are not inputs to the audit.

Accordingly, omitted confounding, geographic shift, leakage, and correlated-feature regimes remain false permissions in all 30 seeds, with temporal drift permitted in 29/30.

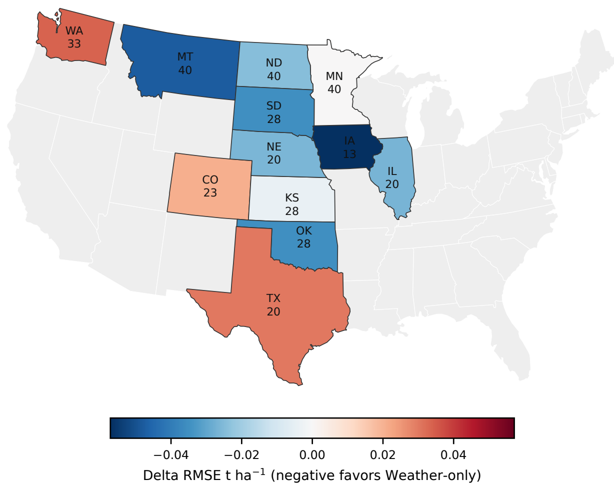

Fig. 4. State-level heterogeneity on the locked 2016–2025 crop panel. Colors show Weather-only minus zero-residual ΔRMSE; negative values favor Weather-only. Labels report state abbreviation and locked-row count. Gray states are outside the study panel. Subgroup estimates are descriptive and not separate module decisions.

TABLE III
SYNTHETIC SCENARIO-LEVEL CALIBRATION. GT IS EVALUATION-ONLY; THE RULE USES ONLY OBSERVABLE MODULES A, B, AND E ACROSS 30 SEEDS.

Scenario	GT	Claim	A	B	E	Rule
Corr. feat.	no	event	100%	100%	100%	100%
Geo. shift	no	event	100%	100%	100%	100%
Imbalanced tail	yes	event	100%	100%	100%	100%
Leakage	no	event	100%	100%	100%	100%
Meas. error	no	event	70%	100%	83%	67%
Moderate signal	yes	event	100%	100%	100%	100%
No signal	no	event	0%	47%	10%	0%
Omitted conf.	no	event	100%	100%	100%	100%
Small sample	yes	event	93%	97%	97%	93%
Spatial mismatch	yes	event	43%	100%	80%	40%
Strong signal	yes	event	100%	100%	100%	100%
Temporal drift	no	event	97%	100%	100%	97%
Train-only detr.	yes	event	100%	100%	100%	100%
Weak signal	no	event	10%	80%	53%	7%

B. Agricultural External-Resolution Check

The county case tests whether finer spatial resolution changes the audit verdict. The locked pre-model population contains 574 counties with at least ten training-period observations from 2000–2025, official county yields, static county interior points, and annual NASA POWER weather aggregates. Validation selected a weather-only ExtraTrees residual model from Ridge and ExtraTrees metadata, weather, and full-feature candidates. On the untouched 2022–2025 holdout, its RMSE was 13.47 versus 13.78 bushels per acre for the zero-residual baseline, but the paired year-block 95% CI for the difference was $[-0.67, 0.26]$. Module B, which compares Full with Metadata-only, passes (ΔRMSE $= -0.71$, 95% CI $[-1.23, -0.34]$), but Module A does not pass under the prespecified upper-CI rule. Finer resolution therefore provides incremental feature evidence without overall eligibility, so the allowed outcome remains model-descriptive explanation only.

C. Cross-Domain Sanity-Check Permission Case

Absolute RMSE values are reported in domain-specific units—t ha^{-1} for the state crop panel, bushels per acre for the county case, and MWh for PJM—and are not directly comparable across domains. The cross-domain comparison therefore concerns pass/does-not-pass verdicts under the same audit logic.

The PJM case serves as a positive control: it tests whether the audit permits interpretation when the applicable predictive evidence is strong. Its target is continuous electricity demand, so the crop-specific extreme-event module is not invoked. Public daily PJM electricity-demand and day-ahead forecast records from EIA Form EIA-930 [30] are fit with Calendar-only ExtraTrees features on January–September 2024 and evaluated on locked October–December records; Full adds the day-ahead demand forecast. Module A compares Full with a prespecified train-period mean-demand naive baseline and passes: locked RMSE is 77,521 versus 335,037 MWh, with paired bootstrap ΔRMSE 95% CI $[-296.7, -221.8] \times 10^3\,\mathrm{MWh}$. Module B compares Full with Calendar-only and also passes: Calendar-only RMSE is 227,417 MWh and the Full-minus-Calendar paired 95% CI is $[-186.0, -119.3] \times 10^3\,\mathrm{MWh}$. Because both applicable modules pass, permutation importance may be interpreted as predictive reliance on the locked PJM rows. PJM therefore serves as a positive control showing that the audit can permit interpretation when the applicable evidence is strong.

VI. Discussion

A. Claim Scope and Multiplicity

The synthetic results define the audit's operating range. Locked predictive checks reduce unsupported interpretation when the requested signal is absent or too weak to support the claim. They do not identify every structurally invalid design, because leakage, omitted confounding, proxies, and some distribution shifts can preserve predictive performance. The audit therefore evaluates predictive claim eligibility; design validity remains a separate requirement.

Several analyses are locally favorable. The eight- and ten-year history filters improve paired error on a smaller 313-row population; the 2013–2015 rolling fold passes, whereas the 2007–2009 fold significantly favors the baseline; severe tails improve RMSE and MAE but do not pass rank and top-k recovery; and the county analysis passes Module B while Module A does not pass. The favorable results occur under smaller evaluation populations, non-primary temporal folds, or settings in which another required module still does not pass. They therefore guide future design choices but do not replace the primary locked verdict.

The audit separates primary decisions from descriptive measurements before the locked period is examined. Module A, primary Module B, and the primary $z < -1$ Module E define the below-trend event permission rule; more severe tails, alternative histories, detrending choices, resampling units, stage proxies, crop/state diagnostics, and expanded model families are sensitivity or diagnostic evidence. They clarify robustness and design trade-offs without replacing the primary locked claim.

B. Reproducibility and Traceability

The reproducibility record contains input hashes, target construction, feature availability, forbidden-column checks, locked configuration, seeds, row IDs, output-scale definitions, predictions, bootstrap draws, and table inputs, so paired comparisons can be recomputed on identical locked rows.

C. Limitations and External Validity

State-representative weather can miss within-state exposure, weather covariates are correlated, and the panel lacks soil, irrigation, cultivar, management, and phenology data. The primary interval resamples only ten locked year-block units, and cluster sensitivities encode different dependence assumptions without removing that limitation. This negative result limits the present target, resolution, features, and model family; it does not establish that weather lacks scientific relevance for yield or that a stronger model cannot exist. The synthetic benchmark shows that predictive adequacy and feature-group value cannot identify omitted confounding, leakage, correlated-feature structures, or unobserved distributional violations by themselves. The separately locked county check shows that finer resolution does not automatically rescue overall adequacy, while the PJM permission case demonstrates that the audit permits interpretation when locked signal is strong. Future work needs prospectively versioned inputs, finer exposure measurements, design-stage validity checks, and an independent future test interval.

VII. Conclusion

The claim-eligibility audit requires locked evidence before post-hoc explanations are used to support progressively stronger claims. Synthetic calibration shows both its value and its limit: false permission falls from 240/240 to 171/240 invalid runs, but predictive checks do not identify structural violations that preserve performance. In the agricultural cases, the required modules do not support weather-specific or event-level interpretation; in the PJM case, the applicable modules pass. The audit therefore provides a practical control on claim scope, while leakage, confounding, measurement, and distribution-shift validity remain part of study design.

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